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Use of mobile fast pyrolysis plants to densify biomass and reduce biomass handling costs—A preliminary assessment

Phillip C. Badger*, Peter Fransham

Renewable Oil International® LLC, 3115 Northington Court, Florence, AL 35630, USA

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Abstract

ROI BioOil plants can be made modular and transportable, allowing them to be located close to the source of biomass and the subsequent transportation of high energy density BioOil to a central plant. Conversely, one central BioOil plant could supply several energy users in distributed locations, or several plants could supply numerous end-users, just as in the petroleum industry.

Renewable Oil International® LLC (ROI) is one of several developers of fast pyrolysis technology. The production of BioOil can convert raw biomass into a low-viscosity liquid that, depending on the moisture content of the feedstock, increases the energy density of biomass by a factor of 6 to 7 times over green wood chips. The increase in energy density increases the amount of energy that can be hauled by standard tanker trucks versus a chip trailer van by a factor of two. Capital costs, exclusive of land costs, are comparable for a 50 MW_e biomass handling system at the power plant. Land area requirements for fuel storage and handling are reduced roughly half for BioOil systems versus solid fuel handling systems. No analysis was made of operating and maintenance costs.
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1. Background

The US Forest Service and the Bureau of Land Management are responsible for maintaining forest health—including forest fire management—on lands owned by the US Government. In the last several decades, US Forest Service policy has been to extinguish all forest fires as soon as possible. In the 1990s it became obvious that the prevention of naturally occurring forest fires allowed the forest undergrowth to gradually build up, with the result an increase in the number and size of forest fires—particularly in the Western US. The National Fire Plan [1] was initiated in 2001 in response to the devastating fire season of 2000 and in August 2002 President Bush announced his Healthy Forests Initiative that identifies steps to reduce the risk of catastrophic wildfires and improve the health of the nation's forests [2]. After the 2001 fire season, the US Forest Service switched to mechanical methods for clearing undergrowth. Based on open bids, the cost for mechanical removal in the Western US can range up to $2470 per hectare ($1000 per acre) [3]. In addition to the cost of removal, marketing of the harvested resource is difficult due to the distance to potential markets and the inherent cost of transporting and handling low-density materials.

A significant portion of biomass feedstock costs—especially from forests—can be attributed to the "handling" associated with moving them from their point of production to their point of conversion or end-use [4]. Traditionally, handling includes harvesting, chipping, loading onto trucks, and transportation to their end-use point. Additionally, handling includes the operations at the end-use point including weighing, dumping, screening, grinding, storage, various conveying operations, and metering into the end-use system. Handling solid forms of biomass is expensive for a number of reasons including the number of operations required and the low bulk density of the feedstocks [5].

If solid forms of biomass could be converted into a liquid BioOil (pyrolysis oil), it would simplify handling transportation, storage, and use of biomass. Additionally, as

*Corresponding author. Tel.: +1 256 740 5634; fax: +1 256 740 5635.
E-mail address: pbadger@renewableoil.com (P.C. Badger).

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discussed later in this paper, BioOil has a much greater energy density than raw biomass. The combination of simplified handling and greater energy density significantly reduces the cost of biomass transportation and increases the feasibility for large-scale bioenergy facilities.

Several organizations in the world, including Renewable Oil International® LLC (ROI), are developing fast pyrolysis technology to convert biomass resources into high-quality BioOils[1] [6] that are partially characterized by their low viscosity (similar to No. 2 fuel oil). BioOils can potentially be used in most applications where No. 2 fuel oil is used, including fueling space heaters, furnaces, and boilers (including cofiring in utility boilers); and fueling certain combustion turbines and reciprocating engines, as well as serving as a source of several chemicals. These attributes also allow biomass energy to provide base load or peaking power, something that is typically difficult to achieve from biomass energy. Significant work has already been performed on using BioOils for energy. For example, starting in 1989, a boiler used in Wisconsin was fueled with BioOils for several years [7]. Orenda Aerospace Corporation, a Division of Magellan Aerospace located in Ontario, Canada, has a 2.5 MW_e combustion turbine certified for use with BioOils [8].

The biomass conversion technology used by ROI falls into a class normally referred to as pyrolysis or thermolysis. Pyrolysis by definition is chemical change brought on by heat in the absence of oxygen. Fast pyrolysis processes are characterized by [9]:

- Very high heating and heat transfer rates, which usually requires a finely ground biomass feed.
- Carefully controlled pyrolysis reaction temperature of around 500 °C in the vapor phase, with short vapor residence times of typically less than 2 s.
- Rapid cooling of the pyrolysis vapors to give the bio-oil product.

One goal of ROI is to develop small, transportable BioOil production plants, which in turn would allow the plants to be taken to the source of biomass and raw biomass converted into a liquid BioOil in the field. In the case of grass crops, the goal is to use conventional harvesting equipment to cut, field dry, rake, chop with a forage harvester, and blow into a trailing wagon for transport to a field edge for processing into BioOil. Similarly, the goal for forestry resources is to use conventional feller-bunchers and skidders to harvest trees and bring them to a forestry landing site where they would be chipped and processed into BioOil. Standard whole tree chippers would be used for chipping; however, the thickness of the chips would be adjusted to approximately 1.5–3 mm (1/16–1/8 inch) to facilitate rapid drying and fast pyrolysis processing. This paper focuses on some of the potential advantages on biomass transportation and handling from the use of transportable BioOil processing plants that are taken to the biomass source.

[1]High-quality BioOils made with *fast* pyrolysis processes are differentiated in the industry from other bio-oils by use of the term "BioOil."

2. Biomass densification and hauling comparisons

Low bulk densities characterize raw biomass, which can be as low as 64–96 kg/m^3 (4–6 lb/ft^3) in its loose form [4]. Current technologies to increase biomass bulk density include pelletizing, cubing, and baling. Table 1 compares energy densities for various biomass forms. The ratios in Table 1 show that BioOil has an energy density of 6 to 7 times that of green whole tree chips at 45% moisture content and 56%, respectively. For comparison purposes, the density of a solid block of a low-density wood (Douglas fir) and a solid block of high-density wood (oak) are also shown [10], along with their corresponding energy densities under air-dry conditions. Also shown for comparison purposes are energy densities for cubed biomass and pellets and baled grasses.

Although the energy density of BioOil is an advantage, this advantage is somewhat limited by legal restrictions on truck gross weights. Load limits on major highways in the US are typically 36.3 t (80,000 lb) gross vehicle weight (GVW). In addition to an overall weight limit, each single axle (one axle with a wheel on each end) is limited to 9.1 t (20,000 lb) and each tandem axle (one axle with two wheels on each end) is limited to 15.4 t (34,000 lb). Regardless of these limits, the actual allowable weight depends on the vehicle's GVW rating, and if the vehicle's rating is less than the legal road limit, the GVW becomes the legal limit [11].

As a rule, the tare weight for a tractor–trailer combination is approximately 12.7–13.6 t (28,000–30,000 lb), leaving 22.7–23.6 t (50,000–52,000 lb) for a net load based on the 36.3 t (80,000 lb) legal load limit. Whole tree chips truck vans (Fig. 1) in the US typically haul 24.5–25.4 t (54,000–56,000 lb) per load. For green whole tree chips, using the energy values shown in Table 1, a trailer vanload of green wood chips is equivalent to 210–270 GJ (200–250 million Btu). A large tanker trailer as shown in Fig. 2 can carry 35.6 kl (9500 US gallons) of No. 2 fuel oil; equivalent to 31 t (68,200 lb) GVW. BioOil has a lower heating value of 13–18 MJ kg^{-1} (5590–7740 Btu lb^{-1}) [12], thus based on weight, a standard tanker truck can haul roughly 558 GJ (530 million Btu) per load, or about two times the amount of whole tree chips in a loaded trailer van. Note that GVW, not volume, limits the amount of BioOil that can be carried by the tanker truck. The tanker truck has enough volume to haul 43 t (95,000 lb) of BioOil. Special permits can be obtained for moving heavier loads; however, these permits are for infrequent moves such as required for heavy construction equipment [11]. The operating costs for each trucking system (solid fuel trailer van and tanker trailer) are assumed to be roughly the same and therefore inconsequential to this discussion.

Table 1
Comparison of energy densities for various types of biomass

	Density kg m^{-3}	MC % wb[a]	Energy density		Energy density ratio[c]
			MJ kg^{-1}[b]	GJ m^{-3}	
Loose, uncompacted straw or hay	95	20	15.51	1.489	1/15
Baled grasses	190	20	15.51	2.979	1/8
Green whole tree chips	350	56	8.53	3.003	1/7
Green whole tree chips	350	45	10.66	3.754	1/6
Solid wood, low density (Douglas Fir)	400	12	17.06	6.826	1/3
Cubes (e.g., grasses)	450	8	17.83	7.993	1/3
Pellets	640	10	17.45	11.170	1/2
Solid wood, high density (oak)	865	12	17.06	14.744	2/3
BioOil	1200		18.00	21.610	1

[a]Moisture contents are on wet basis and are assumed values based on authors' experience.
[b]Assumed biomass bone dry higher heating value = 19.36 MJ kg^{-1}.
[c]Ratio of BioOil energy density to other forms of biomass.



Fig. 1. A typical trailer van for hauling wood chips or wood waste.



Fig. 2. A typical large tanker trailer with capacity of 35,600 l (9500 US gallons) for hauling petroleum fuels.

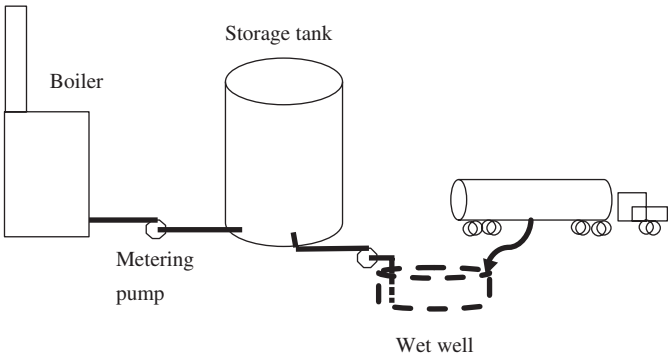


Fig. 3. A BioOil handling system at its end-use point.

3. Handling advantages

Additional savings result from the simplicity associated with handling BioOil at the plant site. As shown in Fig. 3, handling BioOil at the plant site requires a meter (or a scales) for measuring the amount of BioOil unloaded into a wet well before pumping into a storage tank, and a variable speed pump to deliver BioOil to the end-use device. The variable speed pump allows BioOil to be metered to the boiler or other end-use device as required. Storage tanks, like silos for solid fuels, have the smallest footprint for the amount of storage volume. The wet well allows several tanker trucks to dump by gravity at one time.

Contrast Fig. 3 with Fig. 4, which shows a typical industrial biomass solid fuel handling system at a US plant site. This system consists of scales to weigh incoming trucks, one or more truck dumpers depending on the size of the installation, screens and magnets to remove oversize particles and ferrous metals (not shown), a hog to grind oversize particles, some method of storage, some method of reclaiming the fuel, a metering device to feed the conversion device, and conveyor systems in between operations.

Table 2 summarizes the estimated capital costs for a plant-site BioOil handling system and Table 3 summarizes the estimated capital costs for the typical industrial biomass solid fuel handling system at a US plant site. In both tables, the costs are for installed equipment complete with necessary instruments, controls, or ancillary components as required.

Both Tables 2 and 3 assume a 50 MW_e plant size with 28 days of reserve fuel storage onsite and capacity to dump two trucks at once, but do not include land costs. For large plants, the most economical method of storage for raw biomass is on an uncovered earthen pad. One US developer of biopower plants in the range of 10–50 MW_e constructs earthen pads for wood storage with a base of 150 mm (6 inches) crusher run stone, followed by 50 mm (2 inches) of gravel, plus 25 mm (1 inch) of sand sealer leveled over the top and then rolled for compaction. A 150–300 mm (6–12 inch) layer of wood is left on top of soil to prevent scalping of soil and 7.6 m × 7.6 m (25 ft × 25 ft) concrete pads are placed at the truck dumper and feed hopper, where wear from the front-end loader could create problems [5].

The information in Table 3 shows two whole truck dumpers at an installed cost of $360,000 each. For comparison, the wet well system shown in Fig. 3 and Table 2 can unload two tankers simultaneously.

Based on the capital cost estimates in Tables 2 and 3 and excluding land costs, a BioOil handling system is roughly the same cost to construct. A BioOil system, however, requires considerably less land area. A 50 MW_e plant would require approximately 19 million litres (5 million gallons) in storage capacity to provide a 28-day reserve. One 9,375,000 l (2.5 million gallons) welded steel tank with dimensions of 36.6 m (120 ft) diameter and 16.7 m height (35 ft) with an epoxy lining, floating cover, foundation, and installation would cost approximately $1 million [13].

Based on the largest storage liquid storage tank onsite, US Environmental Protection Agency Regulations require at minimum 110% containment around the tank (or tanks) plus freeboard to allow for rainfall [14]. To avoid space confinement issues related to personnel access, containment berm heights are usually limited to 1.2–1.5 m (4–5 ft) [13]. Using a 1.2 m containment berm height, the total area required for two tanks of the same size with 110% containment for one tank would be 1.8 ha (4.5 acres).

In contrast, the storage for a green wood chip pile would require roughly 3.9 ha (9.6 acres), including a 3.7 m (12 ft) driveway around the pile [5]. Part of the reason for the large land area requirement is that the height of woodpiles is limited to 9.25 m (30 ft) by environmental regulations to prevent windblown debris. Also, in practice, it is hard to pile wood higher than 6 m (20 ft) with a front-end loader and there is an increasing safety hazard in using

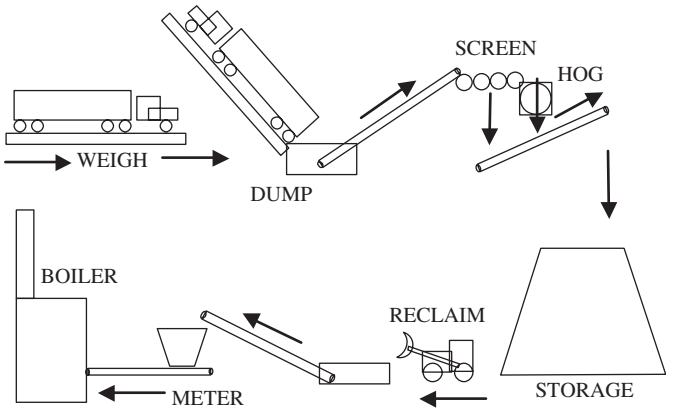


Fig. 4. A typical large-scale industrial wood fuel handling system in the US.

Table 3
Estimated installed capital costs for a typical industrial wood chip handling system at a 50 MW_e power plant

	Cost	Ref.
Scales (2 at $110,000)	$220,000	[5]
Bar code scanner & computer system	$10,000	[5]
Whole truck dumpers (2 at $360,000)	$720,000	[5]
Hopper and drag chain live bottom	$28,000	[5]
Magnet and metal detector	$27,000	[5]
Disc screen	$20,600	[5]
Hog	$26,000	[5]
Super structure for hog and disc screen	$13,500	[5]
Conveyor to storage	$35,500	[5]
Earthen pad for outdoor storage, 21.5 ha (9.6 acres)	$256,000	[5]
Concrete pads with pad (2 at $3000)	$6,000	[5]
Front-end loader (2 at $250,000)	$500,000	[5]
Hopper and conveyor to metering bin	$28,000	[5]
Live bottom metering bins	$110,000	Authors' estimate
Total	$2,000,600	

Table 2
Estimated equipment costs including installation costs for a BioOil handling system at a 50 MW_e power plant

	Cost	Ref.
Wet well receiving tank, 75,000 L (20,000 gal)	$100,000	Authors' estimates
Pump 1, wet well to storage tank (2 at $12,000 each)	$24,000	[13]
Pump 2, storage tank to end-use device (2 at $12,000 each)	$24,000	[13]
Hose, pipe, fittings with installation	$20,000	Authors' estimates
9.4 million L (2.5 million gal) tanks, installed (2 at $1,000,000 each)	$2,000,000	[13]
Total	$2,168,000	

wheeled-type front-end loaders to pile wood at heights above 6 m (20 ft) [5].

4. Other factors

No analysis of operating costs for the two onsite handling systems was conducted; however, the BioOil system has significantly less moving equipment and labor associated with it which would provide lower operating and maintenance costs. BioOil handling systems can provide a greater amount of storage at a lesser cost. Furthermore, there are no dust emissions and relatively no noise associated with BioOil handling operations, and since BioOil is stored in a tank, there should be no contamination of the fuel, no fire control or spontaneous combustion problems, and no rainfall runoff control or treatment required. Relative to solid fuels, BioOils are relatively easy to meter and use and have fewer emissions (especially particulates) [8].

5. Potential impact of the mobile fast pyrolysis plants

A wood-fired direct combustion or gasification system must be directly coupled to its energy user. This requirement means that all wood must be hauled in its raw form (and low energy density) to the plant site. In contrast, successful development of mobile plants will allow several BioOil plants to be located in the field or forest close to the biomass source and the subsequent transportation of high energy density BioOil to a central plant. Conversely, one central BioOil plant could supply several energy users in distributed locations, or several plants could supply numerous end-users, just as in the petroleum industry.

6. Conclusions

Conversion of biomass to BioOil, especially near the biomass source, can reduce the cost of biomass harvesting and handling. Although the capital costs for equipment for onsite handling systems are comparable, the land area required for storage is significantly lower for BioOil versus wood chips. Operating and maintenance comparisons for onsite storage systems were not conducted; however, there should be significant savings with a BioOil system since there is significantly less equipment, fewer operators required, and fewer moving parts associated with BioOil systems.

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